

Lineage and Release Timing in Correlated Errors of Open-Weight Language Models: A Pair-Level Study

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ABSTRACT When two open-weight language models both answer a question wrongly, how often do they choose the same wrong option, and is that more closely associated with shared ancestry or with release timing? Across 977 models from the Open LLM Leaderboard, with item-level predictions on the Massive Multitask Language Understanding (MMLU) benchmark validated item by item, two models descended from the same declared base model choose the same wrong option 13.3 percentage points more often than unrelated models of equal accuracy and release gap (95% confidence interval 10.8–15.8; mean agreement 42.3%). The association is graded, from 18.5 points for parent–child pairs to 11.5 for distant relatives, appears separately in each large base-model family and in all 57 MMLU subjects, and holds across twelve pre-specified analyses, root-level inference, item-difficulty-aware null models, a chance-adjusted agreement metric and the removal of duplicate and degenerate models. Release proximity, whether of the models themselves or of their base models, shows no association of stable sign once accuracies are controlled; the primary analysis excludes same-month effects larger than 1.1 points. The analysis plan and sample rules were fixed before any pairwise outcome was computed. The study is diagnostic: it does not test ensembles, and the associations are observational in a population dominated by community fine-tunes.

INDEX TERMS Correlated errors, dyadic regression, ensemble learning, Hugging Face, large language models, model lineage, Open LLM Leaderboard

I. INTRODUCTION

WHEN several public large language models (LLMs) are asked the same hard question, they often fail together, and often in the same way. Kim *et al.* [1] document this across hundreds of open and hosted models: when two models are both wrong, they choose the same wrong option far more often than chance. For anyone who combines models—a majority vote, a routing scheme, or a review step that escalates only when models disagree—this matters directly, because agreement between members is informative only if their errors are not shared [2], [3].

Two explanations for shared errors compete. *Lineage*: a model fine-tuned from another inherits its parameters and therefore, plausibly, its blind spots. *Era*: models built in the same period draw on the same data, recipes and base models and may acquire the same mistakes without shared ancestry.

The two are entangled, because a fine-tune is always released after its base. We measure era by *release timing*: the month each model was uploaded to Hugging Face and, in a post-hoc analysis, the month its root base model was released.

This paper is a diagnostic study. It measures how shared wrong answers relate to lineage and release timing in a large population of open-weight models; it does not build or test ensembles, and whether the associations translate into better voting or routing is a separate question. Because shared errors are a property of *pairs* of models, we model agreement on wrong answers between pairs, with lineage and release-month gap as pair-level predictors, in a design whose analysis and sample rules were fixed before any outcome was computed. Decomposing per-model accuracy into family and era variance components, the obvious alternative, is too imprecise with the few families and release quarters real populations

offer (Supplementary Section S1); an earlier version of this work attempted it, and its findings are withdrawn (Supplementary Section S2).

A. RESEARCH QUESTIONS

- 1) **RQ1:** Among open-weight models, is sharing a lineage root associated with choosing the same wrong answer, at equal release gap and accuracy?
- 2) **RQ2:** Is release proximity associated with choosing the same wrong answer, at equal lineage status and accuracy?

B. CONTRIBUTIONS

- 1) **A pre-specified pair-level design.** Agreement on wrong answers is modelled for pairs of models, with dyadic inference for the dependence among pairs and a simulation check of whether lineage and release timing can be separated in the population studied (Section IV).
- 2) **An item-level validation protocol.** Predictions for 977 leaderboard models are checked item by item against the leaderboard’s stored correctness, and two data-quality problems in the leaderboard—models listed under several names, and models that give one answer letter to almost every item—are documented and shown not to drive the results (Section III).
- 3) **Evidence on the two explanations.** Shared lineage is associated with about 13 points more agreement on wrong answers, more for closer relatives; release proximity shows no association of stable sign, bounded to about one point in the primary analysis (Section V).

Fig. 1 outlines the study.

II. RELATED WORK

Correlated errors in LLMs. Kim *et al.* [1] measure pairwise agreement on wrong answers across 349 Hugging Face models and 71 models from the Holistic Evaluation of Language Models (HELM) and regress it on pair characteristics, finding that a shared provider, a shared base architecture and similar size predict higher agreement, and that larger and more accurate models have more correlated errors. Their Hugging Face regressions include per-model “generation” covariates with small positive coefficients, and they interpret the accuracy result as newer models converging, but they do not measure how close two models’ releases are or separate that from lineage, and the paper does not describe any correction for the dependence among pairs that share a model. Chen [4] shows across 67 models that the gain from routing, voting or mixture-of-agents is capped by the rate at which all models fail the same query. Kuai *et al.* [5] audit 18 models from six families for synchronized failures and reweight verifier ensembles accordingly. Goel *et al.* [6] propose chance-adjusted probabilistic agreement (CAPA), a similarity metric based on overlapping mistakes, and report that mistakes become more similar as capability increases. Kim [7] audits majority-vote gain across subsets of 30 models and finds that, once capability is controlled, shared errors are the only stable predictor

of lower gain. In vision-language models, Bugaud [8] finds correlated mistakes within architectural families and proposes family-aware aggregation. Evidence on families is not uniform: across 20 instruction-tuned models from 10 families, Li and Hai [9] find no association between same-family membership and error dependence once capability is controlled, and Hossain *et al.* [10] report, for four separately pretrained base models, that overlap in one class of errors tracks shared pretraining recency more than architectural lineage. In both, “family” groups separately pretrained models; in our data, related models share a checkpoint. Tieman and Markou [11] show that diversity in how models generate outputs predicts lower correlated failure.

Null models for excess agreement. Jo, Garg and Raghavan [12] show that whether models agree “more than expected” depends on the null model of independence—accounting for item difficulty changes conclusions—and on the population of models and items. Three estimands are in use: correlated correctness (whether two models fail the same items), conditional agreement on the wrong option (whether two models that both fail choose the same distractor [1]) and accuracy-adjusted similarity [6]. Our primary estimand is the second and our secondary the first. Our quantity of interest is a *contrast* between related and unrelated pairs rather than the level of excess agreement, so a null that shifts all pairs equally leaves it unchanged; we test this with item-difficulty-aware nulls.

Lineage and ancestry. Laufer, Oderinwale and Kleinberg [13] map the family trees of about 1.86 million Hugging Face models from declared parents and study how metadata traits are inherited. Horwitz, Shul and Hoshen [14] recover model trees from weights, and Wu *et al.* [15] infer relationships from a compact representation of model behaviour. Tamura *et al.* [16] show that declared parent–child relations improve the prediction of benchmark performance, and Choi *et al.* [17] find that attention-head truthfulness scores are preserved from base models to their descendants. Nikolić *et al.* [18] test whether one model is derived from another from output similarity alone. PhyloLM [19] reconstructs phylogenies from output similarity, and Li *et al.* [20] trace the lineage of post-training datasets. None of these measures whether related models make the same errors, item by item.

Statistical tools. Crossed variance-component models estimated by restricted maximum likelihood (REML) [21]–[23] and generalizability theory [24] underlie the per-model analysis in Supplementary Section S1; Messing [25] shows that ignoring design choices in LLM pipelines understates evaluation uncertainty. Pairwise outcomes are dependent, because every model appears in many pairs; we use dyadic cluster-robust variance estimators [26]–[28] and a delete-one-root jackknife [29]. Ensemble gains depend on error diversity [30], and monoculture arguments [31], [32] treat correlated error as a risk.

Our contribution. We do not claim to be the first study of correlated errors in LLMs. Relative to Kim *et al.* [1], the closest work, the outcome is the same, but lineage is resolved

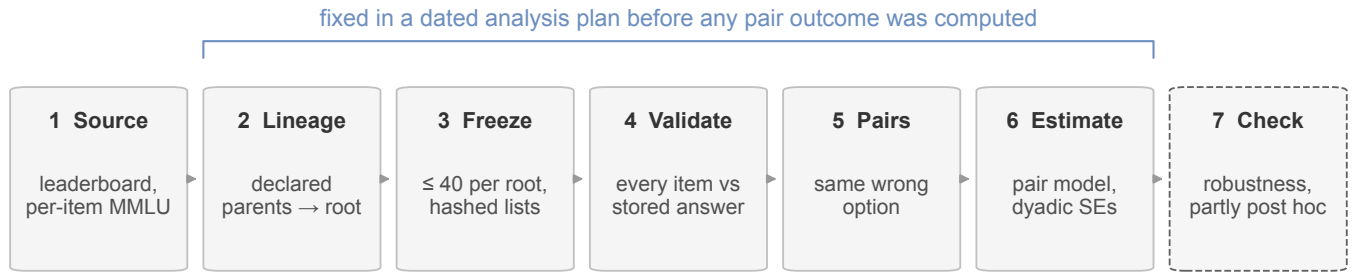


FIGURE 1: Study workflow. Steps 2–6 were fixed in a dated analysis plan before any pair outcome was computed (Section IV-F); step 7 combines pre-specified, exploratory and post-hoc checks. Counts at each step are in Fig. 2.

per model from declared fine-tune ancestry rather than coded as shared provider or architecture, release timing is tested jointly with it at equal accuracy, and inference allows for the dependence among pairs. To our knowledge this is the first study to examine declared ancestry and release proximity jointly as correlates of agreement on wrong answers. TABLE I summarizes the differences.

III. DATA AND LINEAGE

A. SOURCE AND VALIDATION

The first version of the Open LLM Leaderboard [33] evaluated thousands of open-weight models on MMLU [34] (5-shot) with the LM Evaluation Harness [35] and published per-item details, including per-option log-likelihoods. For each model we select the most complete run covering all 57 subjects and read, per item, the per-option log-likelihoods, the correct option, the leaderboard’s stored correctness and, where present, an item hash. A model is accepted only if (i) all 14,042 items are present, (ii) the option with the highest log-likelihood reproduces the stored correctness on *every* item, and (iii) item order and correct options match a reference model. Check (ii) is the decisive one: an earlier version of this work recorded the first answer option for every item, producing constant predictions that no aggregate statistic flagged (Supplementary Section S2), and that error, a misaligned item order and a misread answer key each produce thousands of disagreements. It cannot detect an error in the leaderboard’s own evaluation, such as an unsuitable prompt format for a particular model. For three spot-checked models, extracted accuracy also equals the leaderboard’s official aggregate (43.80, 70.06 and 75.56). Three storage layouts occur among the accepted models (692 in the newest layout, 253 and 32 in two older ones). In the newest the correct-option column is blank and the correct option is taken from the reference model by position, which check (ii) then validates: a misaligned answer key fails on roughly three quarters of items. We keep each model’s chosen option per item; the per-option scores serve only for validation.

Algorithm 1 Lineage root of a model

```

1:  $n \leftarrow$  canonical Hub identifier of the model (renames collapsed);  $d \leftarrow 0$ 
2: while  $d < 20$ :
3:   if  $n$  declares several parents: return unresolved (merge)
4:    $p \leftarrow$  declared parent of  $n$  (expanded population only: at  $d = 0$ , the configuration link if the card declares none)
5:   if  $d > 0$  and  $n$  no longer exists: return unresolved (missing ancestor)
6:   if  $p$  is none and  $n$  is a listed re-upload:  $p \leftarrow$  the checkpoint it copies
7:   if  $p$  is none: if the leaderboard lists  $n$  as a fine-tune, return unresolved (undeclared parent); else return root  $n$ 
8:   if  $p$  was already visited: return unresolved (cycle)
9:    $n \leftarrow$  canonical identifier of  $p$ ;  $d \leftarrow d + 1$ 
10: return unresolved (chain too deep)

```

B. LINEAGE AND RELEASE TIMING

Lineage is resolved by following each model’s declared `base_model` on Hugging Face to a *root*, collapsing repository renames and a documented list of straight re-uploads and quantisations. A chain is unresolved if it passes through a merge, reaches a deleted ancestor, or ends at a fine-tune with no declared parent. Algorithm 1 gives the exact rules. Two populations are defined: *primary*, in which every link comes from a model card, and *expanded*, which also accepts a model’s own first link from the `_name_or_path` field of its configuration file when that field names an existing repository. A *strict* variant of each keeps only chains ending at a checkpoint the leaderboard lists as pretrained.

Declared lineage is sparse. Of the 6,896 models on the leaderboard, 721 are eligible for the primary population. The reasons for exclusion overlap: 4,843 models declare no parent, 1,637 are merges and 776 more descend through one, 1,209 are no longer on the Hub, 385 are adapter or delta weights, 339 chains end at a fine-tune with no declared parent, and 56 reach a deleted ancestor. For the 2,978 models excluded only for lacking a declared parent, the configuration file names an existing repository in 1,073 cases; in the others it names a local path (838), a string that is not a repository identifier (353), the model itself (228), a repository that no longer exists (111) or a merge tool (63), or the field or file is missing (312). Of the 1,073, 574 resolve to an eligible root

TABLE I: Relation to the closest prior work.

Study	Outcome	Relatedness of models	Release timing
Kim <i>et al.</i> [1]	Same wrong option when both wrong	Shared provider, base architecture	Per-model generation covariate; proximity not tested
Goel <i>et al.</i> [6]	Chance-adjusted agreement (CAPA)	–	Capability, not time
Chen [4]	Rate at which all models fail an item	–	–
Kuai <i>et al.</i> [5]	Synchronized failures of LLM judges	Model family	–
Jo <i>et al.</i> [12]	Excess agreement under alternative nulls	–	–
PhyloLM [19], LLM DNA [15]	Output similarity used to infer relations	Inferred from outputs	–
This study	Same wrong option when both wrong; phi; CAPA	Declared ancestry: shared root, tree distance	Upload and root release months, jointly with lineage

TABLE II: Analysed samples. S2: models with accuracy ≥ 0.30 . Agreement: probability of the same wrong option when both models are wrong, Eq. (1).

	Primary	Primary, S2	Expanded	Expanded, S2
Models	590	343	928	589
Lineage roots (with ≥ 2 models)	351 (63)	163 (32)	422 (108)	221 (57)
Upload months	27	18	27	19
Pairs	173,755	58,653	430,128	173,166
Same-root pairs	2,020	1,896	4,836	4,050
Accuracy, median [range]	0.415 [0.229, 0.810]	0.610 [0.302, 0.810]	0.455 [0.229, 0.810]	0.584 [0.301, 0.810]
Jointly wrong items per pair, median	4,109	3,709	4,250	3,820
Agreement a_{ij} , mean	0.423	0.551	0.434	0.547
same-root pairs	0.731	0.739	0.692	0.736
different-root pairs	0.420	0.545	0.431	0.542

and join the expanded population (1,295 models). *Upload month* is the Hub creation month of a model's repository (27 months, March 2022 to July 2024); *root release month* is that of its root.

C. SAMPLES

Each population was sampled with at most 40 models per root and a fixed seed, and the samples were frozen with SHA-256 hashes before any prediction was downloaded (Fig. 2). Of the 1,018 models in the two overlapping samples, 977 passed validation; 41 were excluded and logged, and none was replaced. The analysed primary sample has 590 models from 351 roots (63 with two or more models) and 173,755 pairs; the expanded sample has 928 models from 422 roots (108) and 430,128 pairs. The largest roots are Llama-3-8B and Mistral-7B-v0.1 (39 and 37 primary models). Median accuracy is 0.415, and 247 primary models score below 0.30, close to chance for four options. TABLE II describes the four analysed samples.

D. DATA QUALITY

Two problems in the leaderboard data affect pairwise analyses. First, some models are listed more than once, a redundancy also noted by Paquin and Jain [36]: in the primary sample, 50 leaderboard entries resolve on the Hub to only 23 repositories (renamed or moved), and most such pairs give identical predictions on every item; 193 primary pairs agree on at least 99.9% of items. Second, 21 primary models choose one answer letter on at least 95% of items; all score below 0.30. Both inflate agreement for some pairs, in opposite directions for the lineage contrast. Neither was anticipated in the analysis plan; Section V-E reports the results without them.

E. HOW LINEAGE AND RELEASE TIMING OVERLAP

Fig. 3 places the primary models on a grid of root by upload month and shows what the data can identify. Most roots contribute one model (288 of 351); such models enter only different-root pairs, so the lineage contrast is identified by the 63 roots with two or more models. The fine-tunes of a root are uploaded within a few months of one another, so same-root pairs are concentrated at short gaps (711, 1,004, 259, 45 and 1 pairs at 0, 1–2, 3–5, 6–11 and 12 or more months), and gaps of a year or more are populated almost entirely by different-root pairs.

IV. STUDY DESIGN

A. WHY PAIRS RATHER THAN MODELS

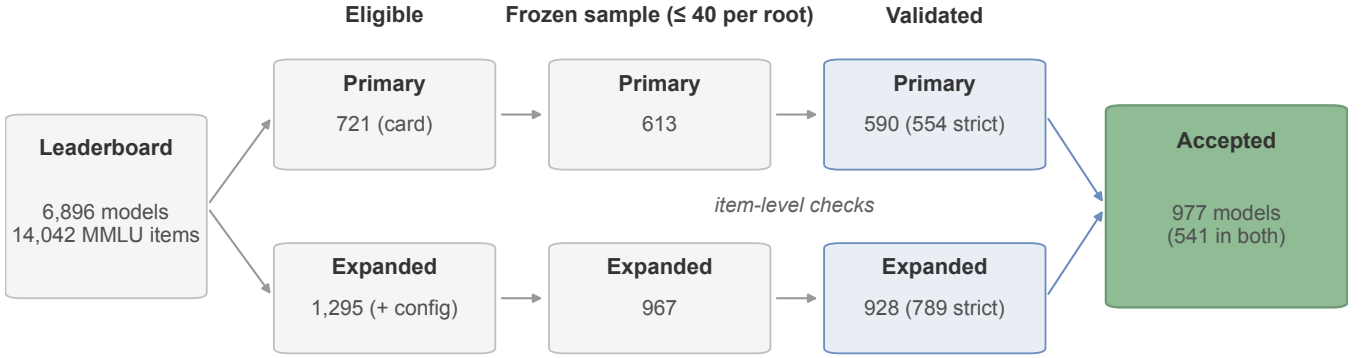
The natural first design treats each model's accuracy as the outcome and decomposes its variance into a family component and a release-era component with a crossed random-effects model. The components are identified in our populations, but they cannot be estimated precisely. Fig. 4 shows the worst-case Monte Carlo root-mean-square error (RMSE) of either variance share across four scenarios: designs with five to seven families, as real populations of pretrained bases provide, have an RMSE of at least 0.18, and bringing both shares within 0.10 requires about 60 families and 14 release quarters (Supplementary Section S1). The constraint is the number of levels of each factor, which more models do not supply. Pairs avoid it: shared errors are a pair-level quantity, lineage and release gap are pair-level attributes, and 590 models yield 173,755 pairs whose comparisons span every combination of lineage status and gap that the data contain.

B. OUTCOME

For models i and j , let c_{ik} be model i 's chosen option on item k and t_k the correct option. The outcome is the conditional agreement [1]

$$a_{ij} = \frac{\sum_k \mathbf{1}[c_{ik} = c_{jk} \neq t_k]}{\sum_k \mathbf{1}[c_{ik} \neq t_k] \mathbf{1}[c_{jk} \neq t_k]}, \quad (1)$$

the probability of choosing the same wrong option given that both are wrong. Every pair contributes all items on which both models are wrong: between 1,166 and 10,820 items (median 4,109). Each item has three wrong options, so two models choosing among them independently and uniformly would give $a_{ij} = 1/3$. The outcome measures *how* two models fail



Excluded and logged, none replaced: 33 no complete run, 7 repository missing, 1 unreadable (of 1,018 frozen)

FIGURE 2: Populations, samples and validation. The primary population accepts only lineage links declared on model cards; the expanded population also accepts a model's first link from its configuration file. The two frozen samples overlap, so the 977 accepted models are counted once. Strict: root verified as a pretrained checkpoint. Item-level checks: all 14,042 items present; the highest-scoring option reproduces the stored correctness on every item; item order and correct options match a reference model.

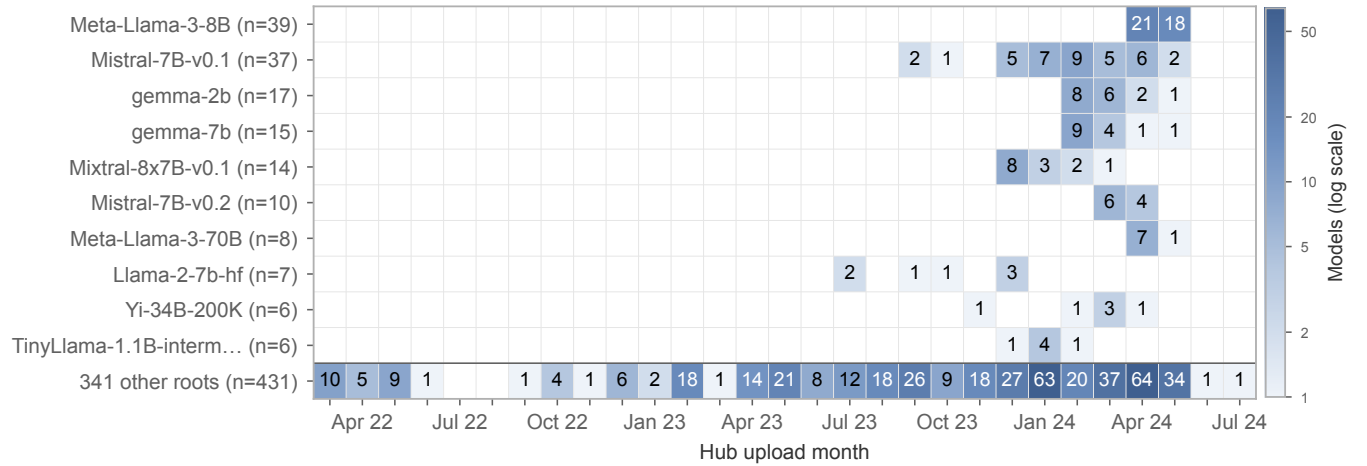


FIGURE 3: Lineage roots by upload month (primary sample, 590 models). Each cell counts the models of one root uploaded in one month (log color scale). The 10 largest roots are shown individually; the bottom row pools the other 341 roots, most of which contribute one model. Fine-tunes of a root cluster within a few months, so same-root pairs occur mainly at short release gaps.

when they both fail; *whether* they fail on the same items is measured by the secondary outcome, the phi coefficient of their error indicators.

C. PAIR MODEL

The pre-specified model is

$$a_{ij} = b_0 + b_L s_{ij} + \sum_h b_{E,h} \mathbf{1}[|m_i - m_j| \in h] + \gamma_1(\text{acc}_i + \text{acc}_j) + \gamma_2|\text{acc}_i - \text{acc}_j| + \varepsilon_{ij}, \quad (2)$$

estimated by ordinary least squares (OLS), where s_{ij} indicates a shared lineage root, m_i is the upload month, and h ranges over gap bins of 0, 1–2, 3–5 and 6–11 months, with 12 or more months as reference. We call b_L the *shared-root coefficient* and $b_{E,0}$ the *same-month coefficient*: the difference in agreement between pairs uploaded in the same month and pairs uploaded 12 or more months apart, at equal lineage status and accuracies. Coefficients are differences in probability;

in the text we report them in percentage points. Both are conditional associations, not causal effects. Fig. 5 illustrates the two predictors: every pair carries one value of each, so same-root pairs occur at several gaps and same-month pairs occur with and without a shared root.

The accuracy terms are pre-specified because two models of similar accuracy tend to fail on similar items, and the items strong models fail are the hardest, which may have more attractive distractors. They are not neutral: if lineage or release timing affects agreement partly through accuracy, adjustment removes that part, so we also report the model without them.

D. SECONDARY OUTCOMES AND NULL MODELS

Five further quantities are used, each as the outcome of Eq. (2) in place of a_{ij} . With $w_{ik} = \mathbf{1}[c_{ik} \neq t_k]$ the error indicator of model i on item k :

TABLE III: Status of each analysis. Pre-spec.: in the analysis plan. Amend.: dated amendment before any pair outcome. Explor.: not in the plan, run with or after the pre-specified analyses. Post hoc: added in response to review. WLS: weighted least squares; CAPA: chance-adjusted agreement.

Analysis	Status
Pair model, Eq. (2); model-level dyadic SEs; phi outcome	Pre-spec.
Simulation check and its criteria	Pre-spec.
Leaderboard version 1; populations; cap of 40	Amend. 1–2
S1 (option distribution), S2 (accuracy ≥ 0.30)	Amend. 3
Root-level SEs; WLS; option-chance outcome; size controls	Explor.
Without accuracy terms; item-difficulty nulls; CAPA	Post hoc
Root release month; duplicate, degenerate and config checks	Post hoc
500-replicate simulation audit; cap-15 subset	Post hoc
Heterogeneity by root, capability and subject; lineage detection	Post hoc
Tree-distance predictor (in the plan, run after review)	Post hoc
Replication on leaderboard version 2 (access-gated)	Not done

G. SIMULATION CHECK

Because same-root pairs are concentrated at short gaps (Fig. 3), a release-time effect could leak into the lineage coefficient or vice versa. Before any pair outcome was computed, we simulated item-level choices on the actual roster. Each model answers correctly with probability $\text{logit}^{-1}(\theta_i - d_k)$, with $\theta_i \sim \mathcal{N}(0.3, 1)$ and $d_k \sim \mathcal{N}(0, 1)$; when wrong, it picks a root-specific preferred option with probability λ_L , a month-specific preferred option (persisting from month to month with probability 0.8) with probability λ_E , an option preferred by all models with probability $\lambda_A = 0.15$, and otherwise a uniform wrong option. Eq. (2) with dyadic SEs was refitted to each simulated data set. The criteria were: false rejection under no effect at most 10%; *leakage*, the rejection rate of the wrong coefficient under a single true effect, at most 10%; and power at least 80% at $\lambda = 0.05$. With 20 replicates per scenario, the cap of 40 met all three. With 20 replicates, rejection rates move in steps of five points, so after the results were known we repeated the check with 500 replicates per scenario (Fig. 6; tables in Supplementary Section S3). The null and power criteria are met: false rejection is 5.8–7.8% for the chosen design, and power at $\lambda = 0.05$ is 100% for both coefficients. Leakage into the lineage coefficient is 10.8% (95% CI 8.4–13.8) at the strongest release-time effect, $\lambda_E = 0.10$, and 12.6% in the expanded design, against the 10% criterion; every other leakage rate lies between 3.6 and 7.8%. The leak is a small positive bias: the simulated lineage estimate rises by 1.4×10^{-4} , about 2% of the simulated release-time effect, which applied to the largest observed same-month coefficient implies about 3×10^{-4} on an estimate of 0.133. The audit also found the model-level SEs 4–6% smaller than the simulated spread of the estimates in the primary design, one reason we report root-level intervals. It ranked the caps differently from the 20-replicate check, under which caps of 15 and none had

failed; cap 40 remains the primary design because it was the dated choice.

V. RESULTS

A. THE RAW PATTERN

Fig. 7 shows the finding before any modelling. At every release gap at which both kinds of pair occur, same-root pairs choose the same wrong option on 71–75% of jointly wrong items and different-root pairs on 40–45% (overall 73.1% and 42.0%, mean 42.3%). Among different-root pairs, agreement declines slowly with the gap, from 0.45 in the same month to 0.37 at a year or more. Kim *et al.* report the same mean for a later leaderboard snapshot [1], but on a different question set with mostly ten answer options, so the coincidence is not evidence of validity.

B. PRIMARY RESULT AND PRE-SPECIFIED ANALYSES

In the primary analysis, sharing a lineage root is associated with a 13.3-point higher probability of choosing the same wrong option (jackknife 95% CI 10.8–15.8; model-level CI 11.8–14.8). Two models uploaded in the same month agree 0.4 points more than two uploaded a year or more apart (jackknife CI -0.5 to $+1.3$). Across the twelve pre-specified analyses (TABLE IV, Fig. 8) the shared-root coefficient ranges from 11.7 to 13.4 points. The same-month coefficient is positive when the answer-option distribution is controlled (S1: $+1.0$, CI $+0.05$ to $+1.95$) and negative among models above 0.30 accuracy (S2: -1.5 , model-level CI -2.7 to -0.2 , jackknife CI -3.0 to $+0.1$): its sign is not stable. The other gap bins show the same pattern. In plain terms, when two fine-tunes of one base model both fail an item, they pick the same distractor about three times in four; two unrelated models do so about two times in five.

C. ADJUSTING FOR ACCURACY

Accuracy can mediate both associations: fine-tunes of one root tend to have similar accuracies, and so do models of one period. Accuracy rose steeply over the period (Fig. 9): the median primary model scores 0.26 among models uploaded in 2022–2023 and 0.59 among those uploaded in 2024 (Spearman correlation with upload month 0.41), and models of similar accuracy agree more (in the pre-specified model, agreement rises with the accuracy sum and falls with the accuracy difference). Without the accuracy terms (Fig. 10), the shared-root coefficient is 28.3 points and the release-gap coefficients are graded, from 8.7 points in the same month to 2.9 at 6–11 months. With them, the shared-root coefficient is 13.3 points and every release-gap coefficient is within 0.6 points of zero in both populations. If accuracy is a mediator, the unadjusted coefficients are total associations and the adjusted ones the part not running through accuracy; if accuracy and the predictor share a cause, adjustment removes a confounded component. The data cannot distinguish these readings. Under either, the predictors differ in the same way: once both accuracies are known, release proximity adds

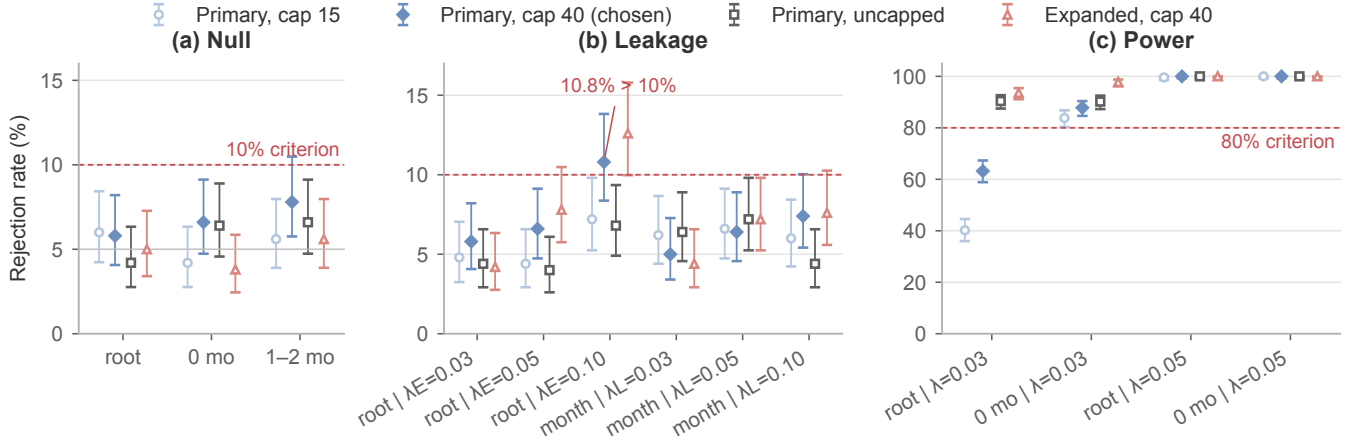


FIGURE 6: Post-hoc audit of the simulation check (500 replicates per scenario; Wilson 95% intervals). (a) False rejection of each coefficient under no effect. (b) Leakage: rejection of the lineage coefficient under release-time-only effects (λ_E), and the larger rejection rate of the 0- and 1–2-month coefficients under lineage-only effects (λ_L). (c) Power. Dashed lines: the pre-specified criteria. The chosen design (primary, cap 40) exceeds the leakage criterion at $\lambda_E = 0.10$.

TABLE IV: Pre-specified analyses. Coefficients of Eq. (2) with 95% CIs from model-level dyadic SEs; same month is the 0-month gap bin relative to 12 or more months, and p refers to it. Strict: root verified as pretrained; S1: answer-option-distribution covariate; S2: models with accuracy ≥ 0.30 .

Analysis	Models	Pairs	Shared root	Same month	p (same month)
Primary	590	173,755	+0.133 [+0.118, +0.148]	+0.0039 [−0.0029, +0.0106]	0.263
Primary, strict	554	153,181	+0.134 [+0.118, +0.149]	+0.0034 [−0.0032, +0.0100]	0.313
Primary + S1 (option distribution)	590	173,755	+0.133 [+0.118, +0.148]	+0.0100 [+0.0005, +0.0195]	0.039
Primary, S2 (acc ≥ 0.30)	343	58,653	+0.120 [+0.105, +0.135]	−0.0145 [−0.0268, −0.0022]	0.021
Primary + S1, S2	343	58,653	+0.117 [+0.102, +0.131]	−0.0124 [−0.0323, +0.0076]	0.225
Primary, strict + S1, S2	319	50,721	+0.118 [+0.103, +0.132]	−0.0138 [−0.0336, +0.0059]	0.170
Expanded	928	430,128	+0.127 [+0.115, +0.140]	+0.0059 [−0.0003, +0.0121]	0.061
Expanded, strict	789	310,866	+0.131 [+0.118, +0.144]	+0.0047 [−0.0016, +0.0109]	0.145
Expanded + S1 (option distribution)	928	430,128	+0.126 [+0.114, +0.138]	+0.0136 [+0.0048, +0.0223]	0.002
Expanded, S2 (acc ≥ 0.30)	589	173,166	+0.122 [+0.111, +0.133]	−0.0089 [−0.0185, +0.0008]	0.071
Expanded + S1, S2	589	173,166	+0.120 [+0.110, +0.130]	−0.0030 [−0.0172, +0.0113]	0.683
Expanded, strict + S1, S2	481	115,440	+0.125 [+0.115, +0.136]	−0.0110 [−0.0261, +0.0042]	0.157

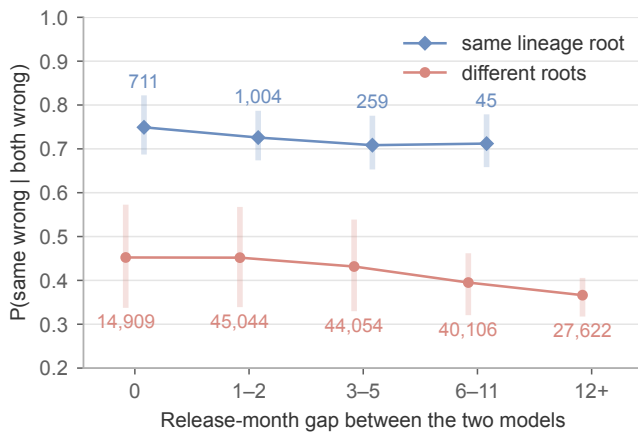


FIGURE 7: Agreement on wrong answers by release-month gap (primary sample, no adjustment). Points: mean of a_{ij} ; shaded bars: interquartile range across pairs; labels: number of pairs. Bins with fewer than 20 pairs are omitted.

nothing of stable sign, while a shared root still adds about 13 points.

D. RELEASE TIMING IN DETAIL

RQ2 asks whether release proximity is associated with agreement at equal lineage and accuracy. Fig. 11 shows every release-gap coefficient under the two timing measures. With upload months, all four gap bins are within 0.6 points of zero in both populations. With the roots' release months—closer to when the base model's data were assembled, and 12,271 different-root primary pairs have roots released in the same month—the coefficients are again near zero: −0.1 points for the same month (CI −1.2 to +1.0) in the primary sample, +0.9 (CI −0.4 to +2.2) in the expanded sample, and negative among models above 0.30 accuracy (−2.2, CI −4.0 to −0.4). Neither measure shows a positive association.

Because this is an absence of evidence, we bound it. The 90% jackknife interval for the pre-specified same-month coefficient excludes associations larger than 1.1 points in the primary sample and 1.3 points in the expanded sample, and 2.0–2.8 points among models above 0.30 accuracy, against 13.3 points for lineage. These bounds were not specified in advance. Release proximity may relate weakly to *which items* fail together: the phi coefficient of error indicators and

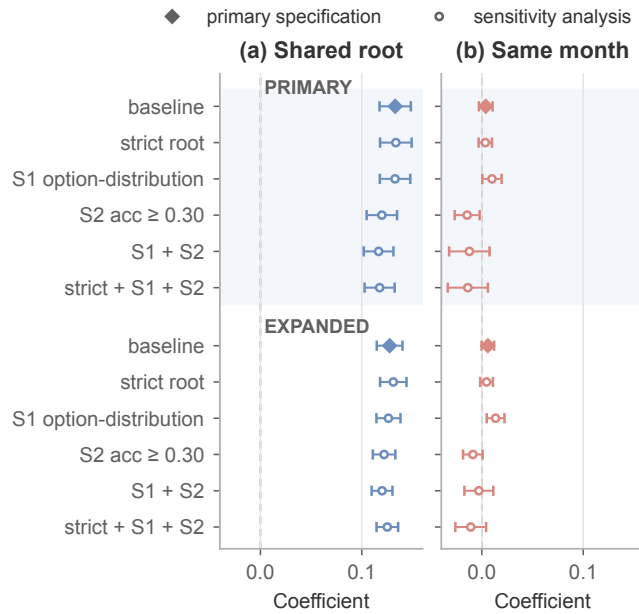


FIGURE 8: Shared-root and same-month coefficients in the twelve pre-specified analyses (95% CIs, model-level dyadic SEs), on one horizontal scale. Diamonds: the primary specification of each population.

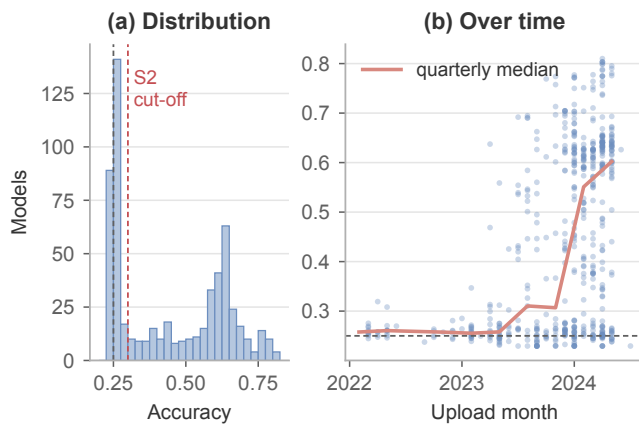


FIGURE 9: Model accuracy (primary sample, 590 models). (a) Distribution; dashed lines mark chance for four options (0.25) and the S2 cut-off (0.30). (b) Accuracy by upload month, with the median of each quarter containing at least 10 models.

a CAPA-style agreement over all items show small positive same-month associations (+1.8 and +1.3 points), and excess co-failure over a Rasch model shows one of 0.36 points per item; the CAPA and Rasch versions disappear among models above 0.30 accuracy. Whatever release-time association exists concerns which items are failed, not which wrong option is chosen.

E. ROBUSTNESS

TABLE V collects every robustness check for the primary sample; detailed tables for all samples are in Supplementary Section S4.

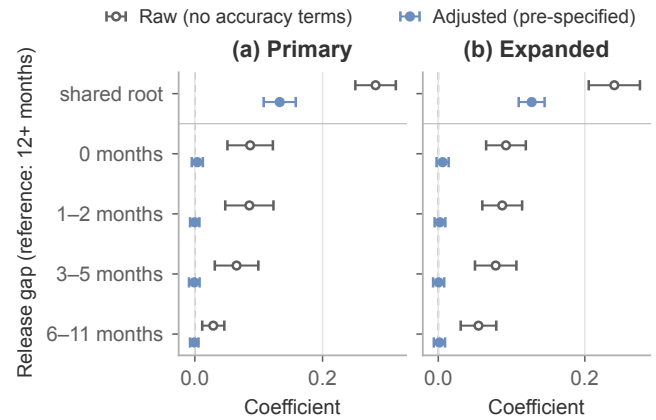


FIGURE 10: Coefficients with and without the accuracy terms (delete-one-root jackknife 95% CIs). Release-gap bins are relative to 12 or more months. Adding the accuracy terms halves the shared-root coefficient and removes the graded release-gap association in both populations.

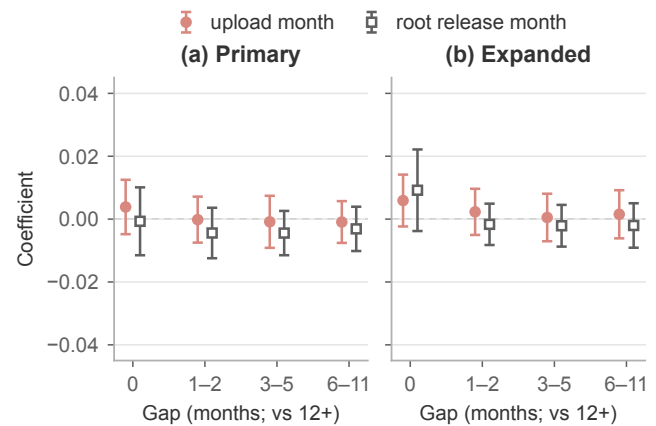


FIGURE 11: Release-gap coefficients under two timing measures (pre-specified model with accuracy terms; delete-one-root jackknife 95% CIs). Upload month: the Hub creation month of each fine-tune. Root release month: that of its lineage root. Each bin is relative to a gap of 12 or more months.

Inference. Fig. 12 compares the three variance estimators in all four samples. Root-level procedures widen the shared-root interval by 10–26% (root-level dyadic) and 39–63% (jackknife), and every interval stays above 0.09. Only the S2 same-month coefficient changes conclusion: its jackknife interval includes zero.

Outcome definition. Weighting pairs by their number of jointly wrong items (weighted least squares, WLS) gives 14.2 points for lineage. Subtracting the agreement expected from each model's distribution over answer options gives 12.5 points. The phi coefficient and the CAPA-style agreement, which also count whether the same items fail, show lineage associations of 17.0 and 19.4 points.

Item-difficulty-aware nulls. With the nulls of Section IV-D, item-level distractor popularity accounts for almost all of the average agreement, which falls from 0.423 to 0.002

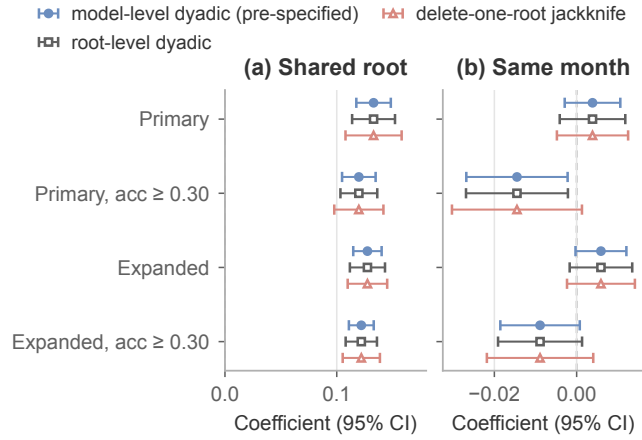


FIGURE 12: Sensitivity of the intervals to the unit of dependence (95% CIs). Model-level dyadic: pairs sharing a model may be correlated (pre-specified). Root-level dyadic: pairs touching a common root may be correlated. Jackknife: one root, and every pair involving it, omitted at a time.

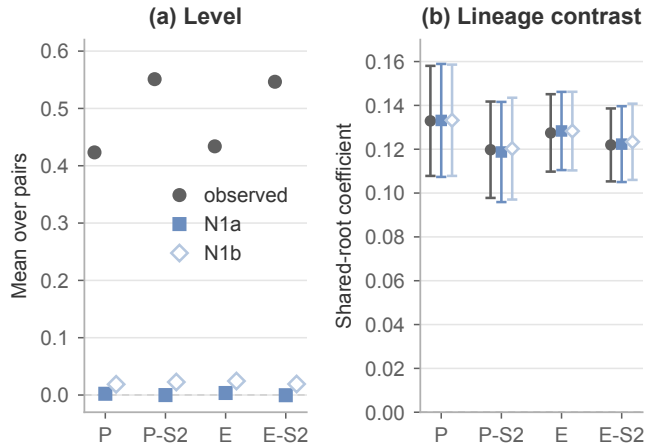


FIGURE 13: Item-difficulty-aware nulls (delete-one-root jackknife 95% CIs). (a) Mean outcome over pairs: observed agreement, and agreement in excess of the item-level distractor null (N1a, all models; N1b, roots weighted equally). (b) Shared-root coefficient for the same three outcomes. P: primary; E: expanded; S2: accuracy ≥ 0.30 .

(N1a) and 0.019 (N1b), consistent with [12]; the lineage contrast is unchanged at 13.3 points (Fig. 13), and changes by less than 0.2 points in every sample. The nulls shift related and unrelated pairs almost equally.

Controls, timing measure and data quality. Adding model size and a same-architecture indicator leaves 12.6 points; lineage and architecture are nested here, so this partitions one association. Measuring release timing by the roots' release months leaves the lineage coefficient at 13.1 points (Section V-D). Removing the degenerate models and collapsing identical-prediction clusters gives 12.8 points; keeping only models whose configuration file agrees with their declared root gives 13.5; the cap-15 subset of the sample gives 14.6.

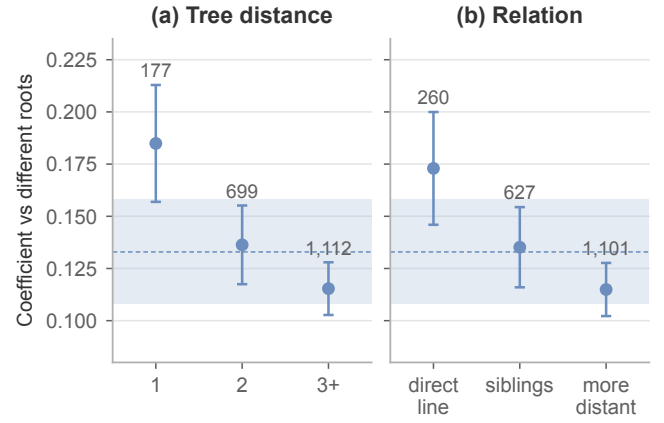


FIGURE 14: Lineage dose-response (primary sample; delete-one-root jackknife 95% CIs; labels: number of pairs). Each coefficient compares one kind of related pair with pairs from different roots, in the pre-specified model with the shared-root indicator replaced. Tree distance: declared links between the two models through their nearest common ancestor. Shaded band and dashed line: the pre-specified shared-root coefficient and its interval.

Independent check of lineage. For 191 primary models, the configuration file names another Hub repository; following that link reaches the declared root in 142 cases (74%), another root in 48 and none in 1. The disagreements mix declaration errors, such as continued-pretraining models declared as their own roots, with configuration files copied from other models.

F. LINEAGE DOSE-RESPONSE

If shared errors are inherited, closer relatives should agree more. Replacing the shared-root indicator by tree distance gives a monotone gradient (TABLE V): 18.5 points for parent-child pairs (177 pairs), 13.6 at distance two (699, mostly siblings) and 11.5 for more distant relatives (1,112). Classified by relation, ancestor-descendant pairs agree 17.3 points more than unrelated pairs, siblings 13.5 and more distant relatives 11.5 (Fig. 14). Closer relatives are also uploaded closer together and often by the same developer, so the gradient is consistent with inheritance but does not isolate it.

G. WHERE THE LINEAGE ASSOCIATION COMES FROM

The contrast is identified by 63 roots, two of which contribute 39 and 37 models, so we examined how it is spread (Fig. 15). By root, each of the seven roots with at least eight models shows its own shared-root coefficient, from 10.3 points (Gemma-7B) to 14.8 (Gemma-2B), every interval excluding zero; the 56 smaller multi-model roots together show 22.0 points, and 19.8 after removing duplicate and degenerate models. That is consistent with the dose-response result, since small roots are often a single developer's successive versions, but we did not test it directly. Dropping any single root, with every pair involving it, leaves the primary estimate between 13.1 and 14.4 points. By capability, the coefficient

TABLE V: Robustness checks (primary sample; 95% CIs from the SE type shown). Same month: 0-month gap bin relative to 12 or more months. ^aDifferent outcome scale (correlation, chance-adjusted agreement or co-failure per item). ^bTree distance replaces the shared-root indicator: the number of declared links between the two models through their nearest common ancestor.

Check	Status	SE	Shared root	Same month
Pre-specified model	Pre-spec.	model	+0.133 [+0.118, +0.148]	+0.0039 [−0.0029, +0.0106]
root-level dyadic SEs	Explor.	root	+0.133 [+0.114, +0.152]	+0.0039 [−0.0041, +0.0118]
delete-one-root jackknife	Explor.	jackknife	+0.133 [+0.108, +0.158]	+0.0039 [−0.0048, +0.0125]
Without accuracy terms	Post hoc	jackknife	+0.283 [+0.251, +0.315]	+0.0866 [+0.0510, +0.1222]
WLS (weight: jointly wrong items)	Explor.	model	+0.142 [+0.125, +0.159]	+0.0117 [+0.0016, +0.0217]
Minus option-distribution chance	Explor.	model	+0.125 [+0.111, +0.140]	−0.0018 [−0.0046, +0.0011]
Phi of error indicators ^a	Pre-spec.	model	+0.170 [+0.149, +0.191]	+0.0184 [+0.0097, +0.0272]
CAPA, chance-adjusted agreement ^a	Post hoc	jackknife	+0.194 [+0.153, +0.234]	+0.0126 [+0.0020, +0.0232]
N1a: minus item distractor null	Post hoc	jackknife	+0.133 [+0.107, +0.159]	+0.0040 [−0.0047, +0.0127]
N1b: as N1a, roots weighted equally	Post hoc	jackknife	+0.133 [+0.108, +0.159]	+0.0038 [−0.0049, +0.0124]
N2: co-failure over Rasch null ^a	Post hoc	jackknife	+0.040 [+0.032, +0.048]	+0.0036 [+0.0012, +0.0060]
+ size and architecture controls	Explor.	model	+0.126 [+0.110, +0.142]	+0.0071 [−0.0027, +0.0169]
Root release month for timing	Post hoc	jackknife	+0.131 [+0.108, +0.154]	−0.0007 [−0.0115, +0.0121]
Without duplicate and degenerate models	Post hoc	jackknife	+0.128 [+0.106, +0.150]	−0.0024 [−0.0081, +0.0034]
Config-consistent lineage only	Post hoc	jackknife	+0.135 [+0.108, +0.161]	+0.0002 [−0.0069, +0.0072]
Cap-15 subset of the sample	Post hoc	jackknife	+0.146 [+0.118, +0.174]	+0.0048 [−0.0045, +0.0141]
Lineage tree distance 1	Post hoc ^b	jackknife	+0.185 [+0.157, +0.213]	—
Lineage tree distance 2	Post hoc ^b	jackknife	+0.136 [+0.117, +0.155]	—
Lineage tree distance 3+	Post hoc ^b	jackknife	+0.115 [+0.103, +0.128]	—

is 13.3 points among pairs whose weaker model scores at least 0.60 and 11.6 at 0.50–0.60; below 0.50 the few same-root pairs give imprecise estimates. *By subject*, the coefficient is positive with an interval excluding zero in all 57 MMLU subjects, with a median of 12.2 points and a range from 7.0 (management) to 18.0 (moral scenarios). The same analysis gives a median same-month coefficient of +0.6 points across subjects, with model-level intervals excluding zero in 13 subjects, 12 of them positive. That is more than the roughly three expected by chance, so a small positive release-proximity association in some subjects cannot be excluded, although model-level intervals tend to be too narrow here.

H. CAN AGREEMENT REVEAL LINEAGE?

If related models share wrong answers, a pair’s agreement should carry information about whether the two share a root. Inferring derivation from output similarity is established [15], [18], [19]; the question here is only how strong that signal is in benchmark error agreement. In the primary sample it is strong: agreement alone separates same-root from different-root pairs with an area under the receiver operating characteristic curve (AUC) of 0.95, and agreement net of the accuracy and release-gap terms with 0.88 (Fig. 16). Above an agreement of 0.7 lie 68% of same-root pairs but only 2.1% of different-root pairs. Because different-root pairs outnumber same-root pairs by 85 to 1, however, most pairs above that threshold are still unrelated (about 3,600 against 1,400), so agreement on one benchmark flags candidate relatives rather than establishing them. The analysis is post hoc and descriptive.

VI. DISCUSSION

What the results say. In this population, which base model a fine-tune descends from is strongly associated with which wrong answers it gives: related pairs agree on wrong answers

about 13 points more often, a 31% increase over the mean, and more the closer the relation. When a model was released adds nothing of stable sign once accuracies are known. This is consistent with Kim *et al.*’s finding that shared provider and architecture predict agreement [1], and refines it: with lineage taken from declared ancestry and timing measured directly, the association runs through lineage.

What they do not say. The association is observational. Fine-tunes of one base share its parameters, its pretraining data and often similar post-training data; the design does not separate these routes or show that fine-tuning causes inherited errors. Nor does it show that release timing is irrelevant: the pretraining era of a base model is shared by its fine-tunes and therefore nested in lineage here, contemporaneous models agree more partly because they are similarly accurate, and timing may relate weakly to which items fail together.

Why lineage and not release timing. A fine-tune starts from its base model’s weights and is typically trained on far less data than its base; much of the base model’s knowledge, and its gaps, plausibly survive. Three features of the results fit this reading: the association grows with closeness of descent, it is present in every MMLU subject rather than concentrated in a few, and it survives nulls that account for which distractors are attractive on each item, so it is not explained by related models simply finding the same items hard. Models uploaded in the same month share only what was common at the time—popular base models, datasets and recipes—and in this population that commonality is already captured by lineage and accuracy. These are hypotheses consistent with the results, not mechanisms the design tests; distinguishing inherited weights from shared fine-tuning data would require models fine-tuned from one base on controlled data.

How far the results generalize. The population is the first Open LLM Leaderboard: open-weight models, mostly community fine-tunes of a few dozen bases released from

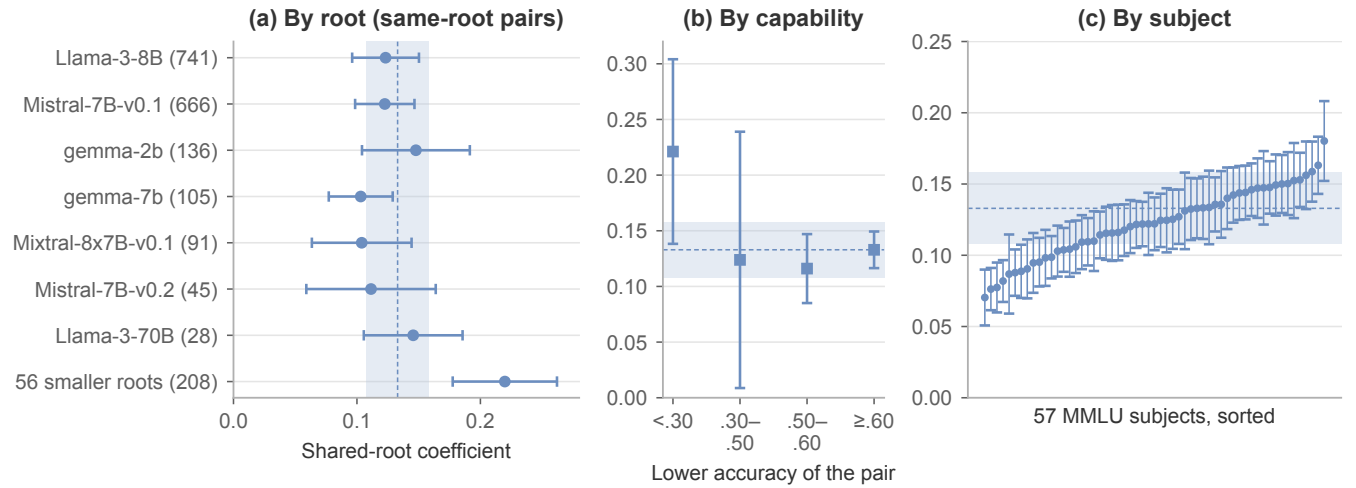


FIGURE 15: Heterogeneity of the lineage association (primary sample; post hoc). (a) Shared-root coefficient for each root with at least eight models and for the remaining multi-model roots pooled (labels: same-root pairs; model-level dyadic 95% CIs, since a jackknife over roots is undefined for a single root). (b) By the lower accuracy of the two models (jackknife 95% CIs). (c) For each of the 57 MMLU subjects, with the outcome computed on that subject's items (model-level 95% CIs). Shaded band and dashed line: the pre-specified estimate and its jackknife interval.

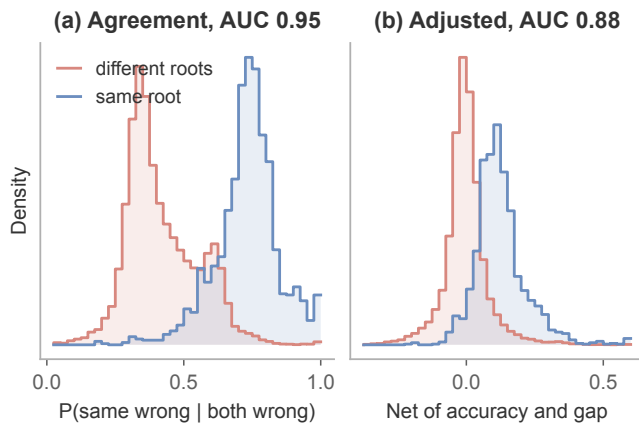


FIGURE 16: Agreement as a signal of shared lineage (primary sample; 2,020 same-root and 171,735 different-root pairs; densities). (a) Observed agreement on wrong answers. (b) Agreement after removing the accuracy and release-gap terms of the pre-specified model. AUC: probability that a random same-root pair scores higher than a random different-root pair.

2022 to 2024, evaluated by log-likelihood on four-option questions. Three features make the lineage result likely to travel: it holds for every large root and every subject, it does not depend on model capability above 0.50 accuracy, and it appears in outcomes that measure which items fail as well as which option is chosen. Two features limit the timing result: pretraining eras are nested in lineage here, and the period spans only 27 months. Closed models, generative tasks scored by judges, and populations of independently pretrained models could all behave differently.

Relation to earlier evidence. Kim *et al.* [1] found that a shared provider and shared base architecture predict agreement; resolving lineage per model shows that, within

providers and architectures, what matters is descent from the same checkpoint, and that the association grows with closeness of descent. Goel *et al.* [6] report that mistakes become more similar as capability rises; our accuracy-sum coefficient is positive, consistent with this, and the unadjusted release-time gradient is what such a capability trend would produce in a population whose models improved over time. Jo *et al.* [12] warn that levels of excess agreement depend on the null; our item-level nulls confirm this for levels and show that the lineage contrast does not depend on it.

What the numbers mean for model selection. When two fine-tunes of one base both fail an item, they choose the same wrong option about three times in four; two models from different roots do so about two times in five. A reviewer or ensemble that treats agreement between two related models as independent confirmation therefore over-counts it more than it would for unrelated models. Release timing, net of accuracy, offers no comparable signal: choosing models from different periods, without changing their bases, would not be expected to reduce agreement on wrong options.

What would change these conclusions. Three findings would weaken them: a lineage contrast that vanishes on other benchmarks or evaluation pipelines; lineage inferred from weights that disagrees with declarations often enough to reverse the dose-response; and, among independently pretrained models released at the same time, a release-time association that the present population cannot show because its pretraining eras are nested in lineage.

Recommendations for leaderboards and model hubs. Three practices would make studies like this one easier and more reliable. Leaderboards could record a canonical repository identifier, since renamed repositories left 50 primary entries pointing to 23 models. They could flag degenerate outputs, such as the 21 primary models that answer one letter

on almost every item. And hubs could encourage machine-readable parent declarations: 70% of leaderboard models declare none, which limits any lineage analysis to a documented minority.

Implications. The results suggest, as a hypothesis for practitioners, that drawing ensemble members from different base models may reduce shared wrong answers more than drawing them from one base at different times. We did not test this. The effect on ensemble accuracy depends on how members are combined [4], and capability-controlled audits find that shared errors predict majority-vote gain only modestly [7].

VII. LIMITATIONS

- 1) **Population and design.** The associations are observational, in a self-selected population dominated by community fine-tunes of a few dozen bases from 2022 to 2024. They may not hold for frontier or closed models, or for separately pretrained models released at the same time.
- 2) **Lineage measurement.** Declared parents can be wrong or missing. Random misclassification would bias the lineage contrast towards zero; misclassification related to how models err could bias it either way. Excluding undeclared models shifts the population towards documented fine-tunes. Estimates agree across the primary, expanded and strict populations (12.7–13.4 points) and in config-consistent models, but lineage was not verified from weights.
- 3) **Release-timing measurement.** Upload and root release months are proxies for when training data were assembled, and for same-root pairs the pretraining era is shared by construction, so pretraining era cannot be separated from lineage.
- 4) **Benchmark and pipeline.** One benchmark with four options is used, and all predictions come from the leaderboard’s evaluation pipeline. Model-specific configuration errors cannot be ruled out; 339 of 977 models score below 0.30, which the S1 and S2 analyses address only in part. Contamination could make agreement partly reflect shared memorization.
- 5) **Outcome scope.** The outcome counts only items both models get wrong; it is not a measure of ensemble performance, and its level depends on the null model and population [12].
- 6) **Identification.** Same-root pairs are almost all within a year of each other, so the data cannot show whether the lineage association persists between fine-tunes released far apart. The contrast comes from 63 roots, two of which contribute 39 and 37 models, although every large root shows it separately and no single root changes it by more than 1.1 points.
- 7) **Design check and plan.** The simulation check narrowly missed its leakage criterion at the strongest simulated release-time effect, and model-level SEs were 4–6% too small in simulation; we therefore report root-level intervals. The analysis plan was not registered ex-

ternally, and several analyses are post hoc (TABLE III).

VIII. CONCLUSION

Among open-weight models on the first Open LLM Leaderboard, two models that share a declared lineage root choose the same wrong MMLU option, when both are wrong, about 13 percentage points more often than unrelated models of equal accuracy and release gap, and closer relatives agree more. The estimate is stable across pre-specified analyses, root-level inference, alternative outcomes, item-difficulty-aware nulls and cleaned samples. Net of accuracy, release proximity—of the models or of their base models—shows no association of stable sign with the choice of wrong option, and in the primary analysis any such association is smaller than about one point. The next step is to test whether lineage diversity improves ensemble, voting or review performance on other benchmarks, evaluation pipelines and independently pretrained models.

DATA AVAILABILITY

Code, frozen model lists with SHA-256 hashes, Hugging Face metadata caches, validation reports and all analysis outputs are available at <https://github.com/Sudharsanselv araj/Identifiable-Variance-Decomposition-of-Correlated-Errors-in-Foundation-Models>; the README lists the commands that regenerate every reported number, table and figure from these outputs. The leaderboard’s per-item details and metadata declare no licence, so the derived item-level answer files and three copied metadata fields are not redistributed; released scripts regenerate them from the public source and verify them against the recorded hashes. The analysis plan and its amendments are in the repository. Supplementary material accompanies this article.

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Use of AI-generated content. The authors used Claude (Anthropic; model Claude Opus 5.5, accessed through Claude Code) [37] in October 2026, with substantial use. It wrote and debugged the analysis, simulation, data-extraction and figure/table-generation code, and drafted the text of every section of this revised manuscript (Sections I–VIII) and of the supplementary material. The authors directed the study, approved each analysis decision and analysis-plan amendment, reviewed and edited the text, and take full responsibility for the content; every reported number is produced by the released code from the committed outputs.

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